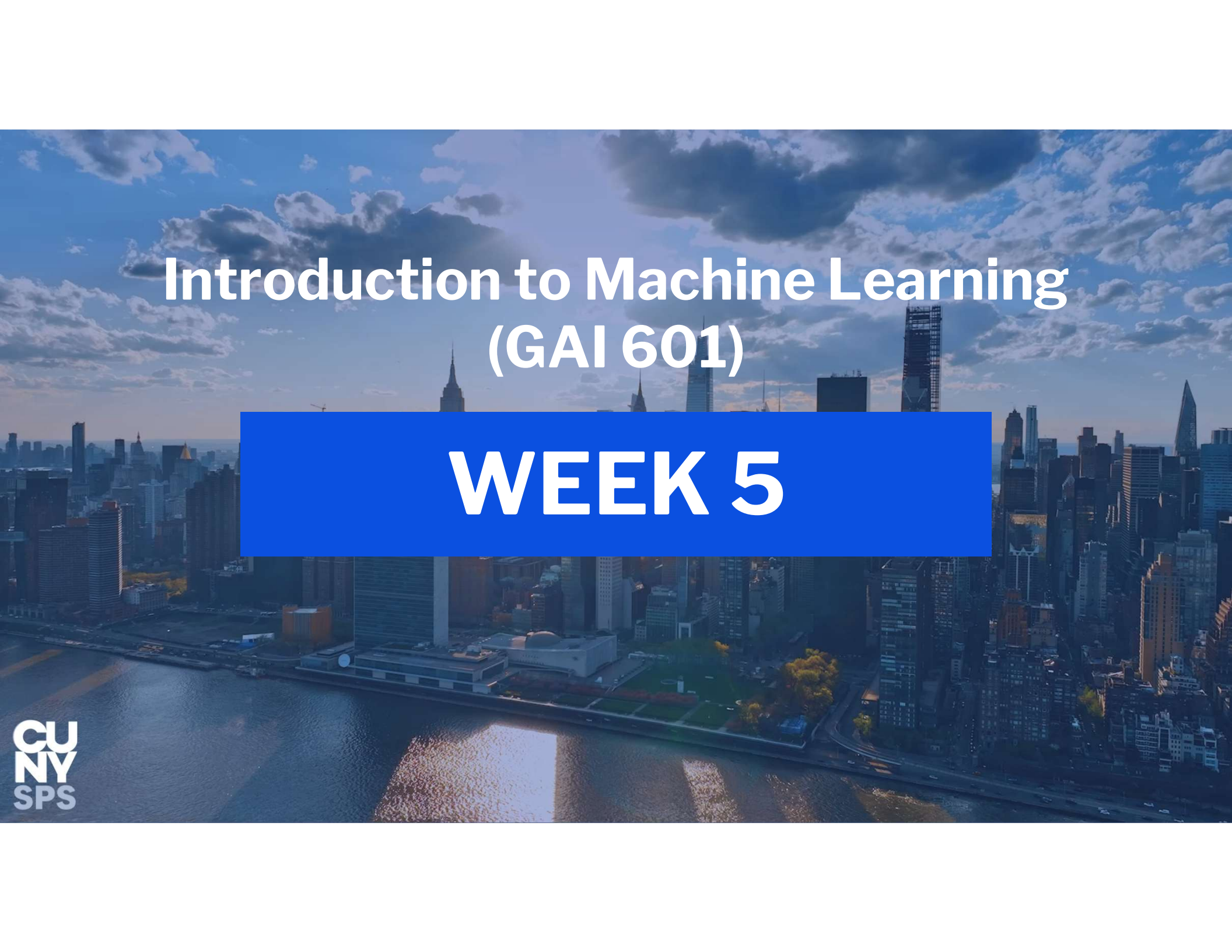


An aerial photograph of the New York City skyline, featuring numerous skyscrapers and the Hudson River in the foreground. A large blue rectangular box is superimposed over the center of the image, containing the text 'WEEK 5'.

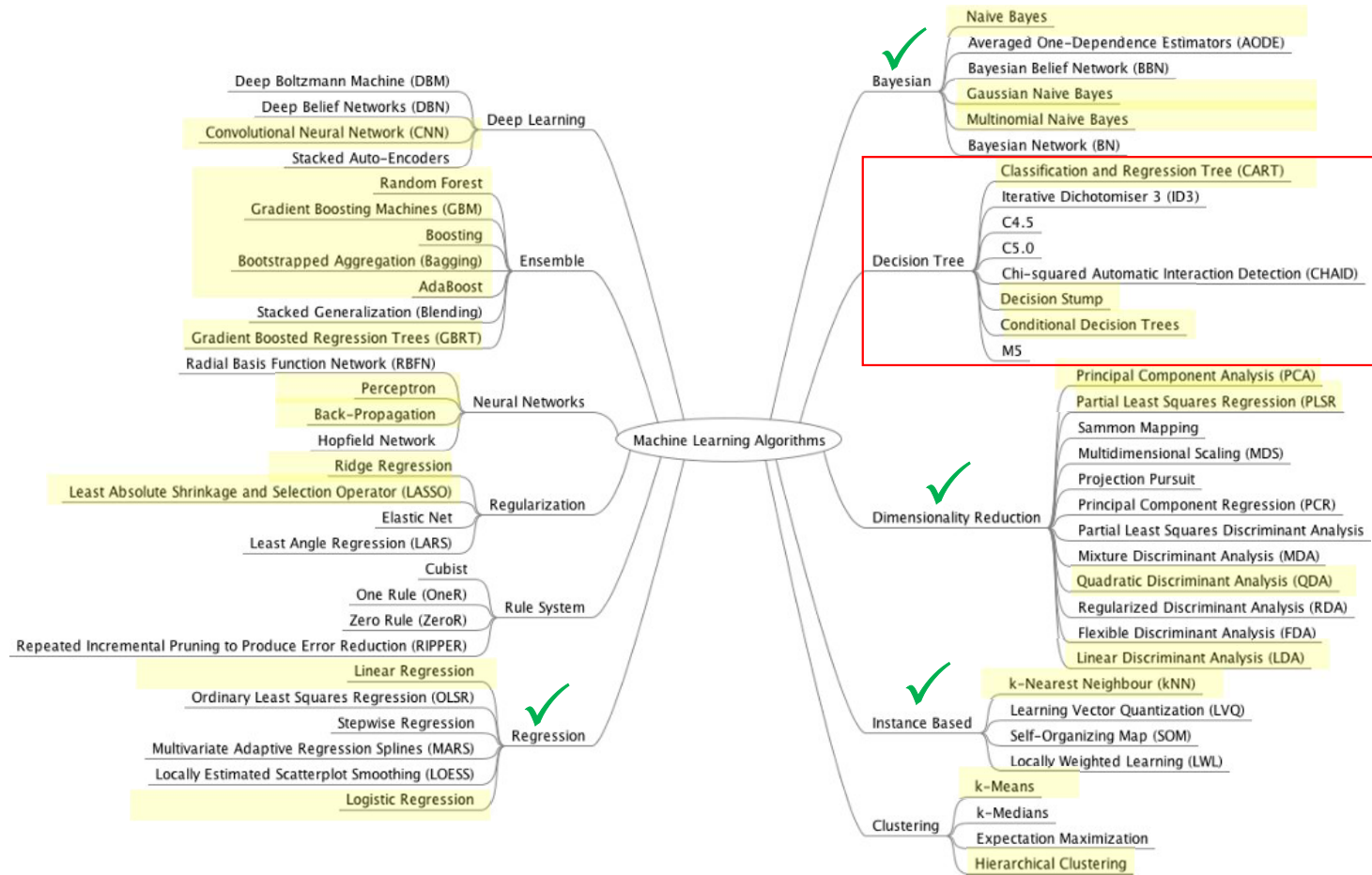
Introduction to Machine Learning (GAI 601)

WEEK 5

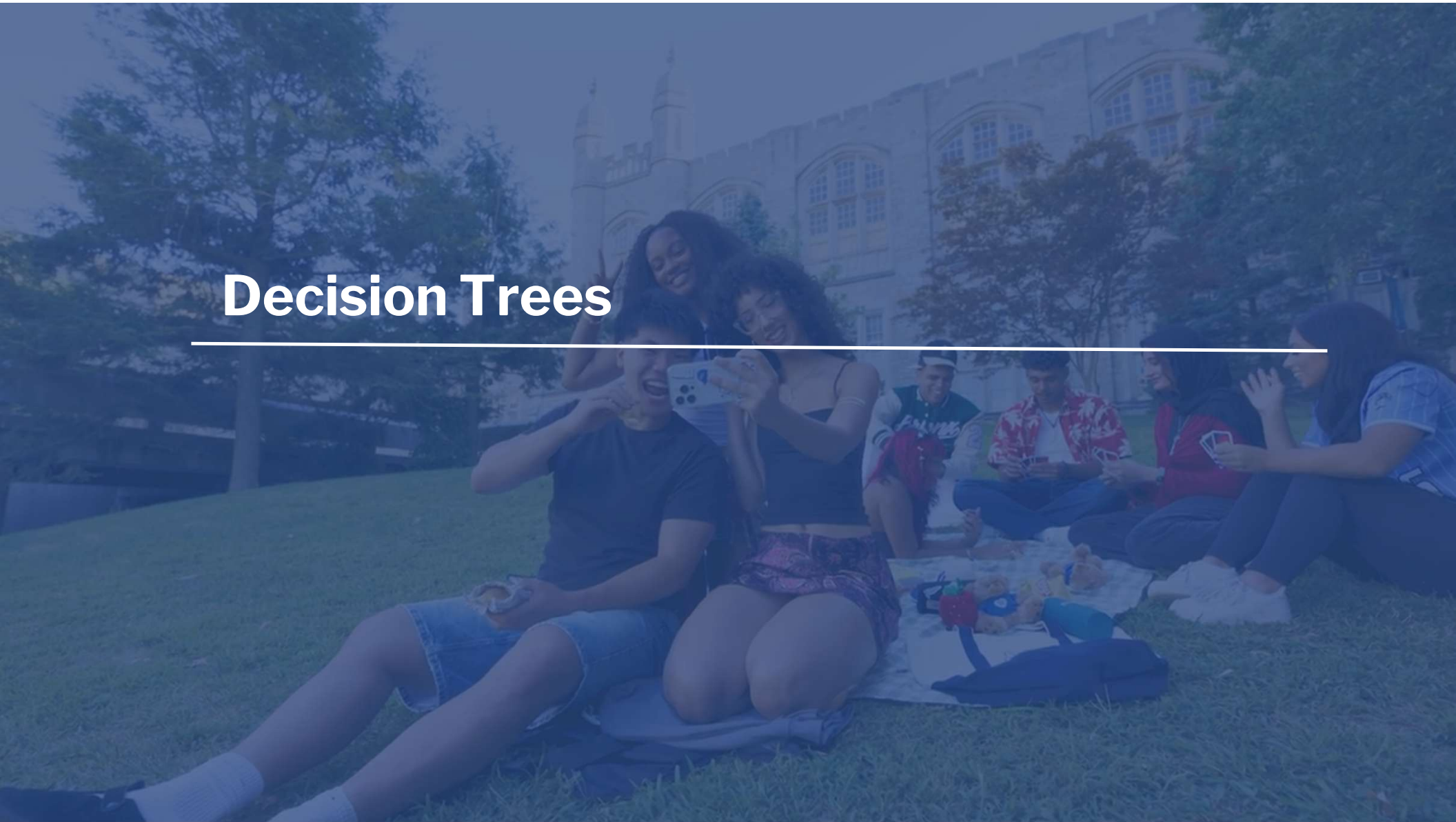
Landscape of Algorithms



Landscape of Algorithms



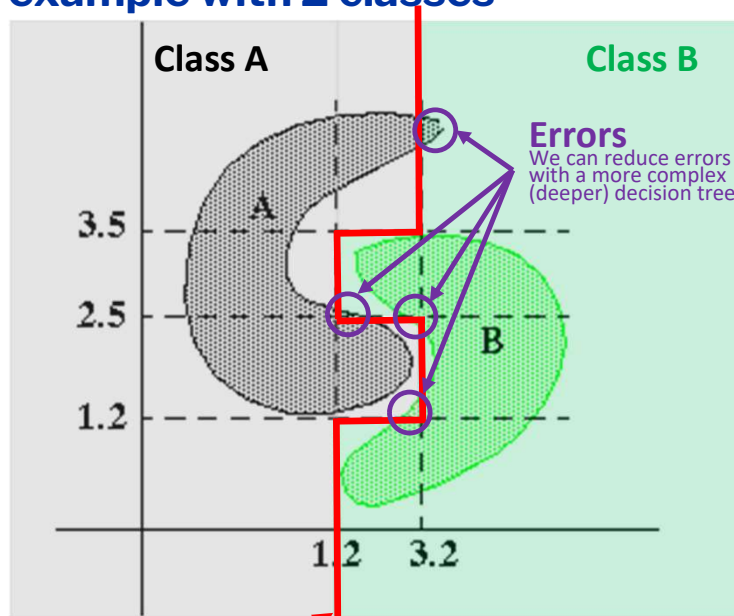
Decision Trees



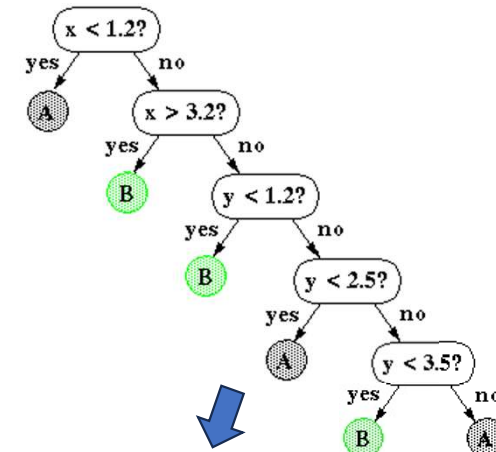
Decision Trees

- Decision trees can produce **nonlinear decision boundaries**
- Can give arbitrarily high levels of precision on the training data

Consider an example with 2 classes



Decision Boundary



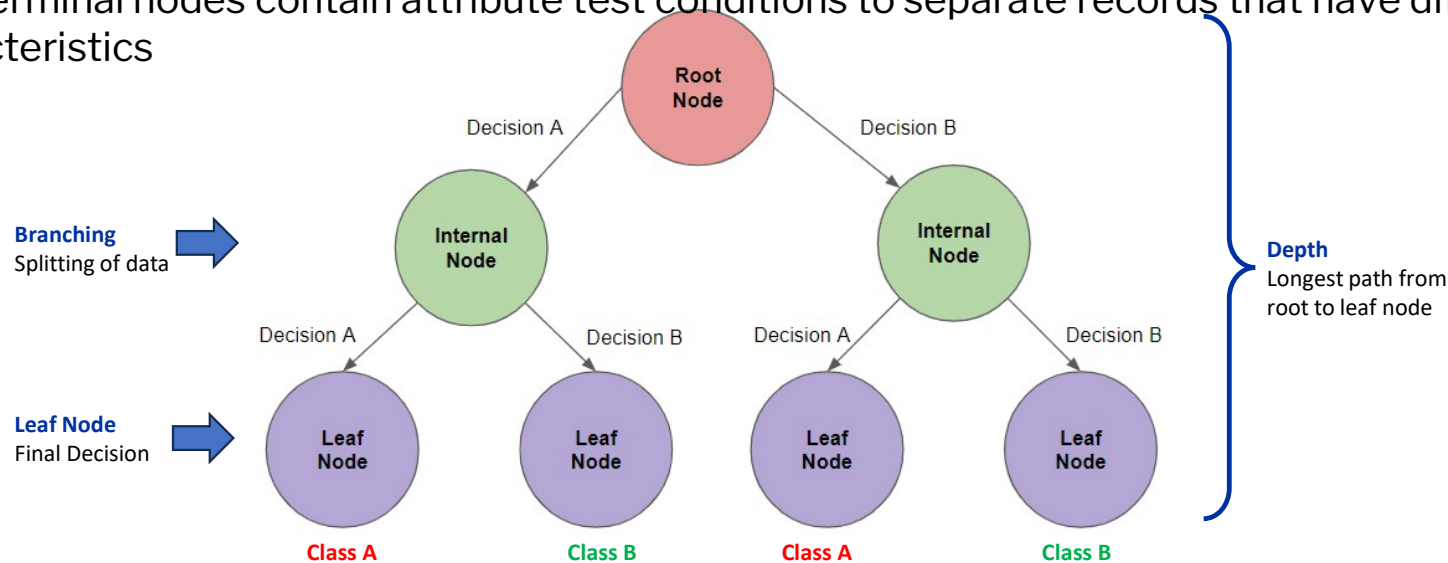
Deterministic

Think of decision trees as simple if...then... else... programs

```
if x < 1.2 then
  output (class=A)
else if x > 3.2 then
  output (class=B)
else if y < 1.2 then
  output (class=B)
else if y < 2.5 then
  output (class=A)
else if y < 3.5 then
  output (class=B)
else
  output (class=A)
```

Decision Tree Nodes

- A decision tree has three types of nodes:
 1. A **root** node that has no incoming edges and zero or more outgoing edges
 2. **Internal nodes**, each of which has exactly one incoming edge and two or more outgoing edges
 3. **Leaf nodes** (or **terminal nodes**), has exactly one incoming edge and no outgoing edges
- Each leaf node is assigned a **class label**
- Non-terminal nodes contain attribute test conditions to separate records that have different characteristics



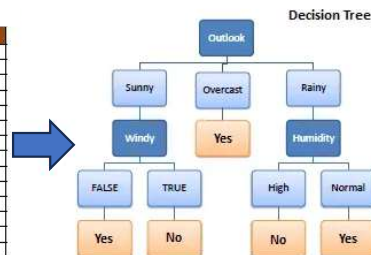
Source: techeastblog.com

Decision Tree: Examples

Playing Golf example

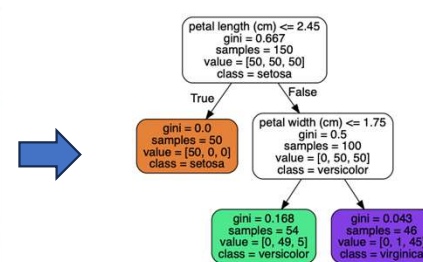
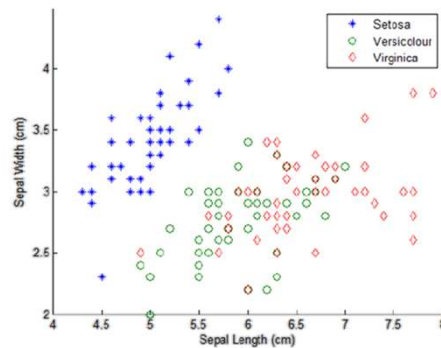
Let's consider the example we looked at last week:

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No



Iris dataset

Let's also look at our favorite dataset:



Source: Evidence Feed Forward Hidden Markov Model: A New Type Of Hidden Markov Model, Michael Del Rose, Wagner Christian, and Frederick Philip, April 2011

Decision Trees: Use-cases

Some examples:

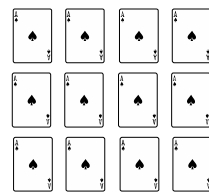
- Fraud detection
- Customer Relationship Management (CRM)
- Medical diagnosis
- Quality Control
- Credit risk management
- Strategy development
- Product Launch
- Recommendation Systems

Generating a Decision Tree

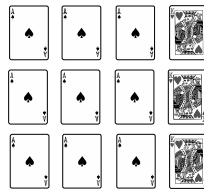
- Referred to as **decision tree induction**
- Exponentially many decision trees can be constructed from a given set of attributes
 - Infeasible to try them all to find the optimal tree
- Usually a **greedy strategy** is employed (top down approach, choosing the optimal split at every branch)
- How do we split the data?

Different “decision tree building” algorithms

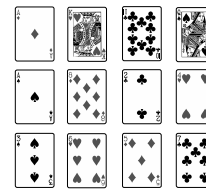
- **Entropy**: Based on randomness of data to be split
- **Information Gain**: How much information a feature provides about a class
- **Gini impurity**: Probability a random data point will be incorrectly classified



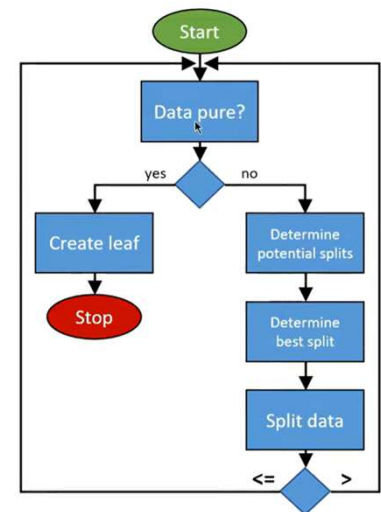
Very pure. Not impure.



Little impure.



Completely impure.



Source: Decision Tree Pruning explained (Pre-Pruning and Post-Pruning), <https://www.youtube.com/watch?v=u4kbPtivVB8>

Decision Trees: Pros & Cons

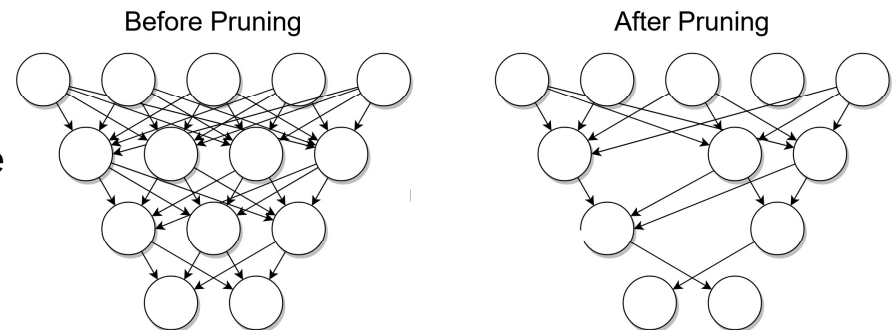
Pros:

- **Interpretable**: Visual. Easy to interpret & explain predicted values
- Handles both Classification and Regression
- **Simple**: Versatile, easy to build and **fast**
- **Non-linear**: Supports complex decision boundaries
- Non-parametric: Has no data assumptions (unlike, for example, Naïve Bayes). Supports both categorical & numerical data)

Cons:

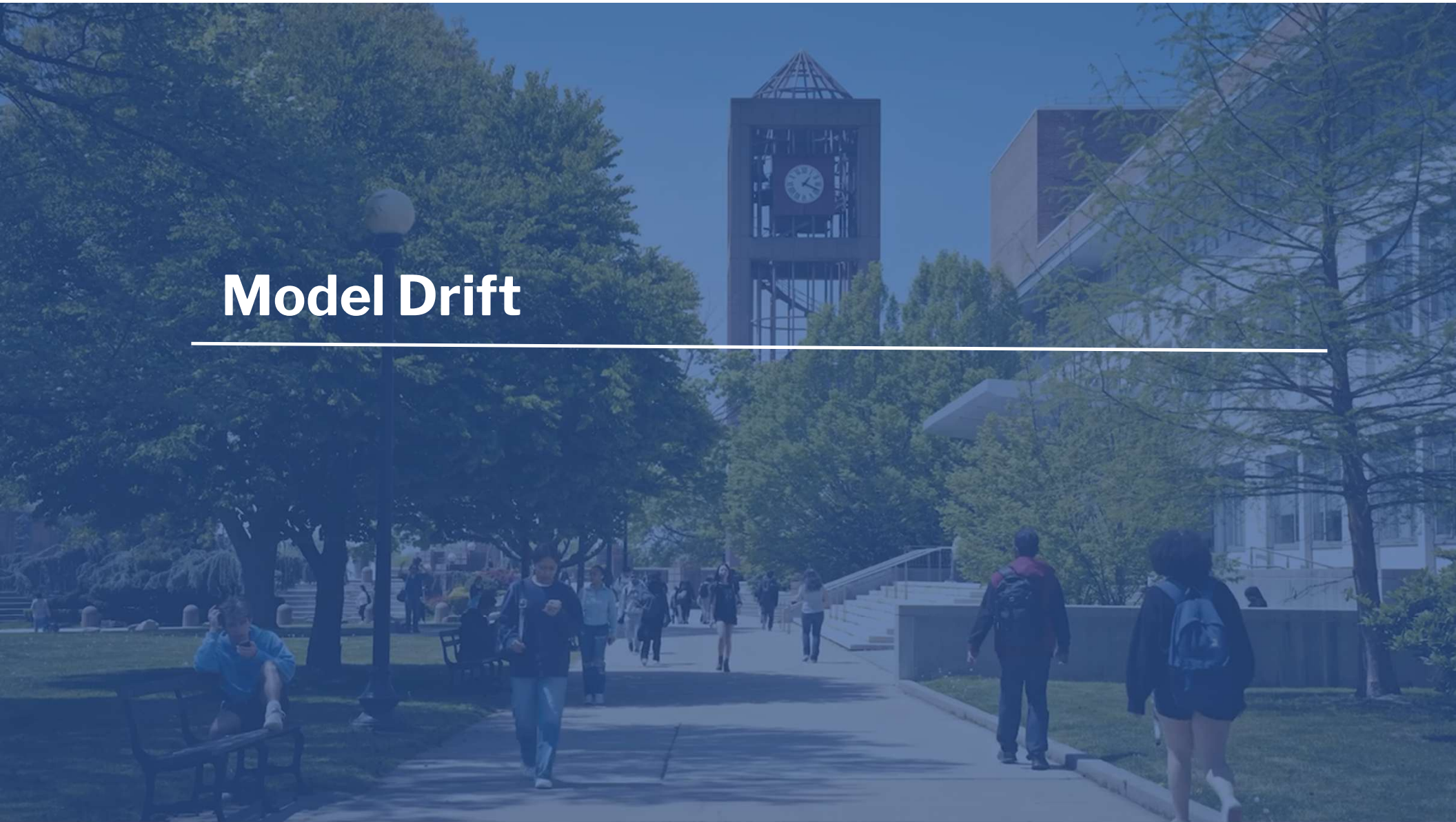
- Variance (deep tree): Susceptible to **overfitting**, where it fits the training data “too well” (under-performing on new data)
- Bias (shallow trees): May favor dominant classes or features
- A **greedy** approach may not find the optimal tree

Source: Wikipedia
May require **pruning** to reduce complexity & overfitting



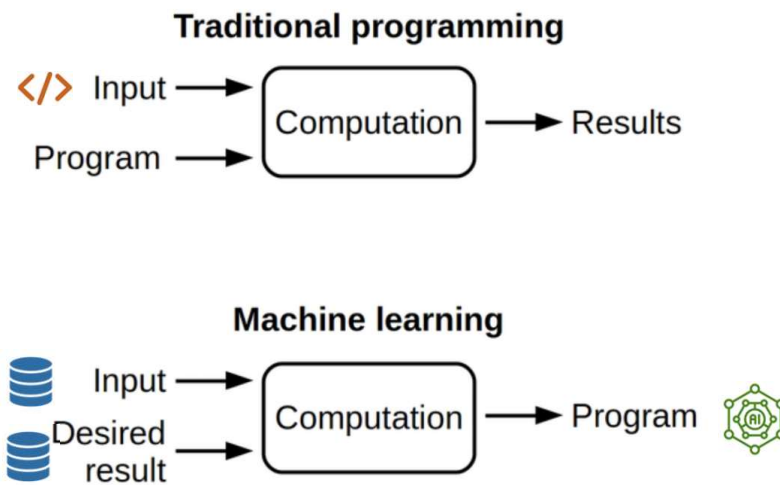
(pruning can be pre- and post-training pruning)

Model Drift



Data is a key input to Machine Learning

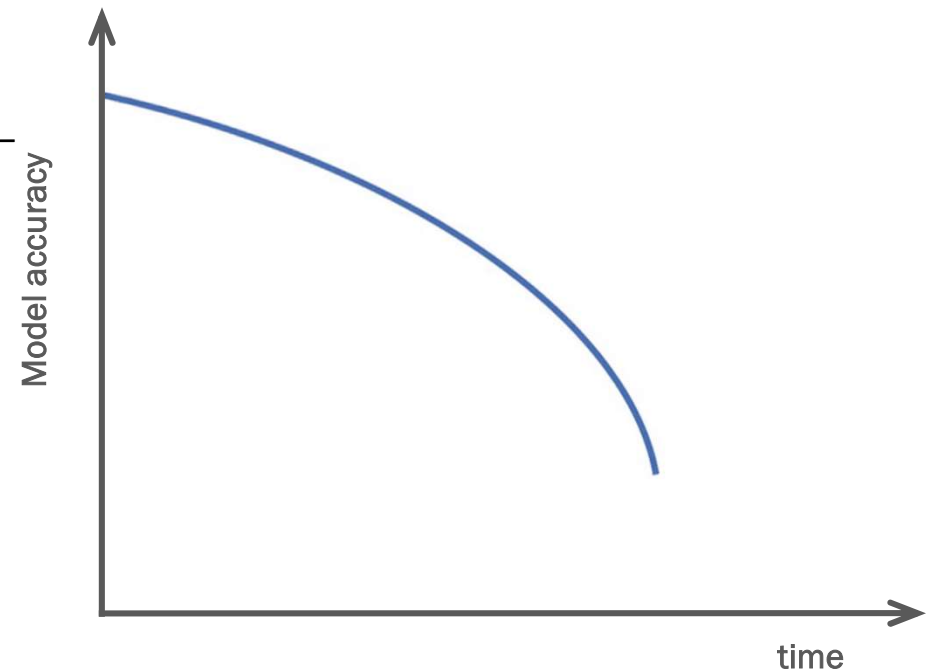
Comparing traditional software development with ML development*



*We'll revisit this diagram in the coming weeks

Real-world data is not static

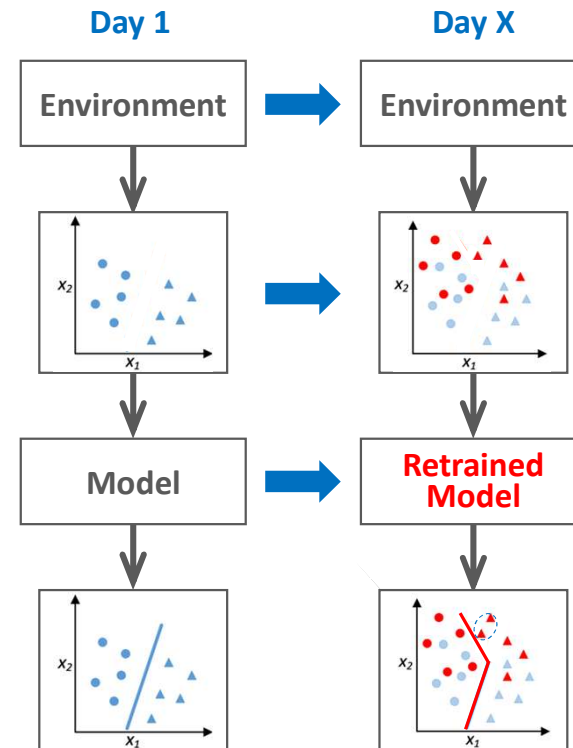
- When a model is put into production model accuracy starts to degrade
- You will never achieve the accuracy in production that you did in training
- The accuracy and quality of models will continue to decay over time
- This is completely different to traditional software – because machine learning is data driven



Source: Wikipedia

Model Decay

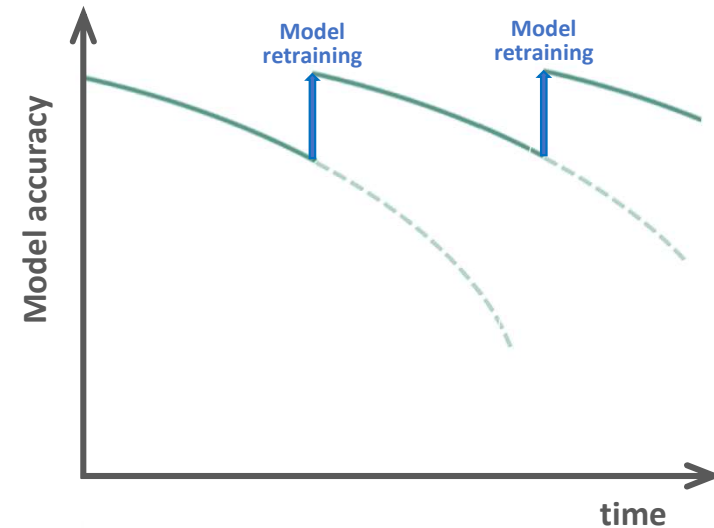
- **Data drift** refers to the changes in the input data (feature dataset) – data is not stationary
- **Concept drift** refers to changes in the environment mechanism (“concept”) that the model is trying to predict
- Concepts – relationships between input & output data - evolve and are not stationary
- **Regime change** refers to a major shift in concept e.g. due to Regulations



Source: Wikipedia

Models Require Maintenance

- Models must be retrained when model accuracy degrades
- Retraining is required for the entire life of the model

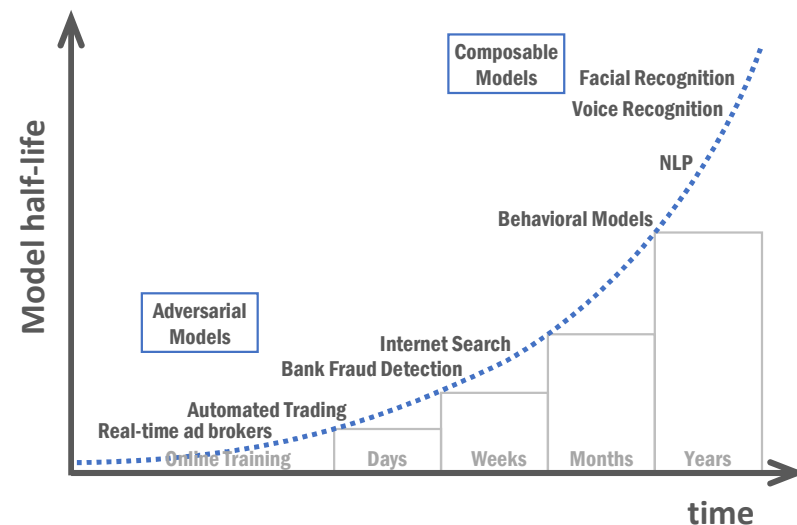


Source: Wikipedia

Models Require Maintenance

When should you retrain your model?

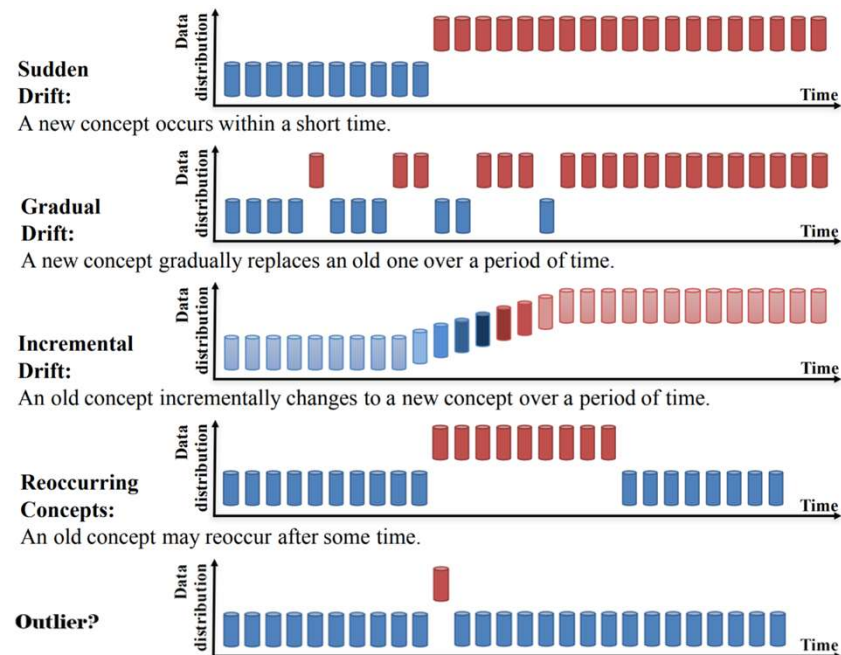
- Models have a gradual decline (the half-life) which determines how long the model is effective i.e. how frequently it must be retrained
- The half-life is determined by how quickly the data distribution is shifting (“data drift”)



Source: Wikipedia

Model Decay

- Another way to think **Concept Drift**: it occurs when the environment changes but the model does not
e.g. COVID-19
- **Periodic drift** reflects periodicity that wasn't captured in training data e.g. seasonality
- **Sudden drift** may also indicate a catastrophic failure in data processing e.g. data pipelines merging data
- To identify drift, it's important to track outliers to identify they are part of a new pattern or true one-off events



Source: Learning under Concept Drift: A Review, Jie Lu et al, 2019

Choluteca Bridge

The Choluteca Bridge was built in 1935 and performed well for 63 years.



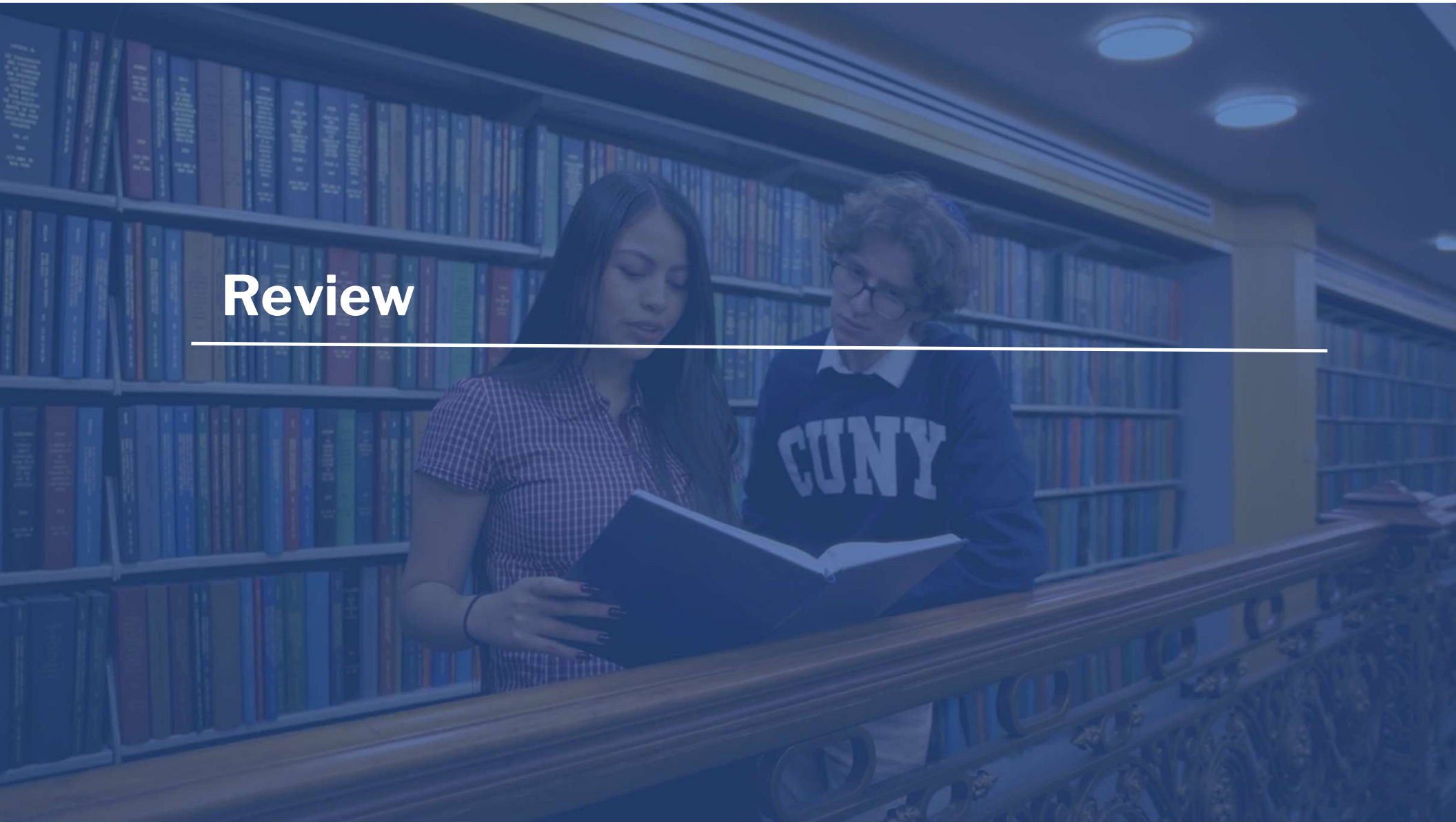
Until Hurricane Mitch in 1998. When it moved the river.



Remember: Even though its called Model Drift, **the only thing that moves is the data.** The model didn't change in the slightest – just like the bridge.

Source: Wikipedia

Review



This week we covered

Lesson Objectives/Topics

- Foundational
 1. Construct decision trees for classification
 2. Visualize and interpret tree splits, nodes, and prediction rules
 3. Evaluate pruning techniques to optimize tree performance and prevent overfitting
 4. Assess the use of decision trees in explainable AI for business applications
- Implementation & Strategy
 - Model drift

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's steel structure and suspension cables are prominent. The water of the harbor is visible in the lower foreground. The sky is clear and blue.

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